Brian Sawaya

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Education

University of California, Berkeley Berkeley, CA
B.S. Mechanical Engineering Expected May 2028
Coursework: Experimentation & Measurements (LabVIEW, DAQ), Dynamic Systems & Feedback, Electronics for IoT

Skills

Design & Analysis: SolidWorks, Onshape, FEA, GD&T, tolerance stack-up, DFM, DFMEA/PFMEA, test protocols, V&V
Test & Build: LabVIEW, DAQ, oscilloscope, strain gauges, 3D printing, machine shop (mill, lathe, laser cutter), soldering
Software: MATLAB, Python, C++, Arduino

Experience

Samtec, *R&D Optics Engineer Co-op* Aug 2026 – Present
- Built a VCSEL test station to recreate component error states tied to field failures and customer issues
- Reverse-engineered a Suruga Seiki motorized positioner's controls and automated VCSEL-to-VCSEL moves in Python
- Designed 3D printed fixtures with kinematic locking and wrote CCD camera software to verify beam alignment
- Validated $\leq$30 µm positioning repeatability with 30-sample tests (standard deviation, commanded vs. actual)

JH Technology Associates LLC, *Systems Engineer Intern* Nov 2024 – May 2026
- Tested and integrated subsystems of a spectrophotometer now on the International Space Station, identifying 35 error states
- Analyzed intensity and absorption data in MATLAB to calibrate the device, increasing signal-to-noise ratio by 20%
- Designed wire management in SolidWorks, soldered components to PCBs, and integrated fluidics

Boeing, *Supply Chain Engineering Intern* May 2026 – Aug 2026
- Redesigned Production Readiness Assessments for 12,000+ suppliers, cutting assessment time ∼40%
- Built AI agents linked to supplier data sources that sorted 6,000+ entries, decreasing turnaround time by 50%
- Joined supplier site visits, documenting process observations used to score suppliers against AS9100 requirements

Stanford University, *Research Intern, Structural Biology* Sep 2024 – Dec 2024
- Designed a skin adapter for an Optical Coherence Tomography scanner used for noninvasive biopsy and cancer monitoring
- Prototyped 7+ CAD revisions with kinematic coupling; reached 4.6 µm Z-repeatability vs. a 500 µm spec

Projects

Industry Client Project (under NDA), *Engineering Solutions @ Berkeley* Sep 2025 – Dec 2025
- Evaluated manufacturability of a proposed assembly process for a medical client; delivered a go/no-go call
- Ran SolidWorks contact FEA to predict part deformation and stress under fixture loads
- Authored the design FMEA and process FMEA, wrote a test protocol, and designed a joint concept in CAD

AutoLab: Automated Cell Culturing Robot, *Project Lead, Neurotech @ Berkeley* Aug 2025 – Present
- Lead a team of 5 on a $2,000 Jacobs Institute grant building a low-cost multi-axis gantry for BSL-2 cell culturing
- Designed a stepper-driven micropipette (A4988 PWM drivers) and a purely mechanical tool changer using kinematic coupling
- Achieved 0.1 mm gantry repeatability; selected to present at a statewide neurotech conference

Diversity By Doing HealthTech, *Intern, Stanford Biodesign* Jan 2025 – Apr 2025
- Led a team of 5 researching endometriosis and wrote a clinical needs statement with Intuitive Surgical mentors

Leadership

MESA, *Ambassador* Feb 2024 – 2025
- Connected 105+ students to internships, raised $2,000 in scholarships, and ran events for 500+ people

Achievements

1st Place, UC Davis Biomedical Engineering Design Challenge (2025) **TEDx Speaker**, *"Engineering Your Luck"* (2025)